



A
Constructive
Small Object
Argument

Paul Seip

Introduction

Motivation

The Main
Theorem

Proof
Strategy

Conclusion

A Constructive Small Object Argument

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Overview

A

Constructive
Small Object
Argument

Paul Seip

Introduction

Motivation

The Main
Theorem

Proof
Strategy

Conclusion

- ➊ Introduction
- ➋ Motivation
- ➌ The Main Theorem
- ➍ Proof Strategy
- ➎ Conclusion



Weak Factorisation Systems

A

Constructive
Small Object
Argument

Paul Seip

Introduction

Motivation

The Main
Theorem

Proof
Strategy

Conclusion

$(\mathcal{L}, \mathcal{R})$ is a *weak factorisation system* on a category $\mathcal{C}$ if

- Every map $f \in \mathcal{C}$ factors as

$$X \xrightarrow[\in \mathcal{L}]{Lf} Ef \xrightarrow[\in \mathcal{R}]{Rf} Y,$$

- $(\dots)$



Examples

A

Constructive
Small Object
Argument

Paul Seip

Introduction

Motivation

The Main
Theorem

Proof
Strategy

Conclusion

- (surjections, injections) is a WFS on the category of sets,
- (injections, surjections) is also a WFS on the category of sets,
Equivalent to the axiom of choice!
- (cofibrations, homotopy equivalences) is a WFS on the category of topological spaces,
- (surjections, injections) is a WFS on the category of groups.

And many more!



Algebraic Weak Factorisation Systems

A

Constructive
Small Object
Argument

Paul Seip

Introduction

Motivation

The Main
Theorem

Proof
Strategy

Conclusion

We can make the notion of a weak factorisation system more ‘algebraic’, the result is called an *algebraic weak factorisation system*.

It consists of a pair of a comonad and a monad (L, R) , its left class is $L\text{-}\mathbf{Coalg}$ and its right class is $R\text{-}\mathbf{Alg}$.

This gives: explicit lifts, explicit factorisations, ...



Small Object Argument

A

Constructive
Small Object
Argument

Paul Seip

Introduction

Motivation

The Main
Theorem

Proof
Strategy

Conclusion

- [Quillen (1967)], small object argument for sets;
- [Garner (2009)], algebraic small object argument for cofibrant generation by a small category;
- [Bourke & Garner (2016)], algebraic small object argument for cofibrant generation by a small double category.

A *constructive* small object argument . . . ?



Constructive Mathematics

A

Constructive
Small Object
Argument

Paul Seip

Introduction

Motivation

The Main
Theorem

Proof
Strategy

Conclusion

Benefits of a constructive proof:

- Has ‘computational content’ (BHK interpretation, Curry–Howard isomorphism);
- Can be internalised in any Grothendieck topos (not just **Sets**).



Constructive Models of Homotopy Type Theory

A

Constructive
Small Object
Argument

Paul Seip

Introduction

Motivation

The Main
Theorem

Proof
Strategy

Conclusion

- Voevodsky's construction of a model of homotopy type theory in simplicial sets.
- Problem: BCP-obstruction (Bezem, Coquand & Parmann), *constructively unprovable* that this is a model of HoTT.
- Solution 1: definition of a *uniform Kan fibration* in cubical sets [Coquand et al. (2015)].
- Solution 2: definition of an *effective Kan fibration* in simplicial sets [van den Berg & Faber (2022)].
- Important step: a proof that these are the right class in an algebraic weak factorisation system.
- We need a constructive small object argument!



The Main Theorem

A

Constructive
Small Object
Argument

Paul Seip

Introduction

Motivation

The Main
Theorem

Proof
Strategy

Conclusion

Theorem (finitary small object argument)

Let $\mathcal{C}$ be a locally small, (countably) cocomplete category, and let $U : \mathbb{J} \rightarrow \mathbf{Sq}(\mathcal{C})$ be a double functor subject to the following conditions

- 1 $\mathbb{J}$ is small,*
- 2 the object Uj is ω -compact for every object $j \in \mathcal{J}_0$.*

Then the AWFS cofibrantly generated by U exists and is finitary.



Step 1: A Beck Theorem for AWFS

A

Constructive
Small Object
Argument

Paul Seip

Introduction

Motivation

The Main
Theorem

Proof
Strategy

Conclusion

Theorem 1 ([Bourke & Garner (2016), Theorem 6])

The functor $(-)\text{-}\mathbf{Alg} : \mathbf{AWFS}_{\text{lax}} \rightarrow \mathbf{DBL}^2$ has as its essential image exactly those concrete double categories $V : \mathbb{A} \rightarrow \mathbf{Sq}(\mathcal{C})$ such that:

- 1 *The functor $V_1 : \mathcal{A}_1 \rightarrow \mathcal{C}^2$ on vertical arrows is strictly monadic;*
- 2 *(...)*



A

Constructive
Small Object
Argument

Paul Seip

Introduction

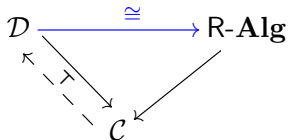
Motivation

The Main
Theorem

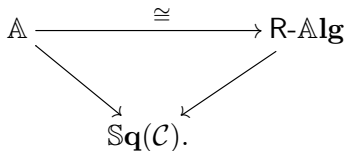
Proof
Strategy

Conclusion

Beck's monadicity theorem:



Beck theorem for AWFS:





Step 2: Construction of the Free Algebra

A

Constructive
Small Object
Argument

Paul Seip

Introduction

Motivation

The Main
Theorem

Proof
Strategy

Conclusion

Theorem 2

Let (T, η) be a pointed endofunctor on a cocomplete category $\mathcal{C}$. If T preserves colimits of ω -chains (in particular if T is finitary), then the free T -algebra exists for every object X in $\mathcal{C}$.

Example: (T, η) is well-pointed.

$$X \xrightarrow{\eta_X} TX \xrightarrow{\eta_{TX}} T^2X \xrightarrow{\eta_{T^2X}} \dots \longrightarrow X_\omega \xrightarrow{\eta_{X_\omega}} TX_\omega,$$

$\nwarrow \quad \text{---} \quad \nearrow$

$$X_\omega = \operatorname{colim}_{n < \omega} T^n X.$$



Step 3: Constructing Colimits of Finitary Monads

A

Constructive
Small Object
Argument

Paul Seip

Introduction

Motivation

The Main
Theorem

Proof
Strategy

Conclusion

Theorem 3

Let $\mathcal{C}$ be cocomplete. Then the category of finitary monads $\mathbf{Mnd}_f(\mathcal{C})$ is cocomplete. Moreover, the functor $(-)\text{-Alg} : \mathbf{Mnd}_f(\mathcal{C})^{\text{op}} \rightarrow \mathbf{Cat} / \mathcal{C}$ creates and preserves limits.

$\lim_{j \in \mathcal{J}} T_j\text{-Alg}$

$T\text{-Alg}$

$\mathcal{C}$



Step 3: Constructing Colimits of Finitary Monads

A

Constructive
Small Object
Argument

Paul Seip

Introduction

Motivation

The Main
Theorem

Proof
Strategy

Conclusion

Theorem 3

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$$\lim_{j \in \mathcal{J}} T_j\text{-Alg} \quad T\text{-Alg} \quad \mathcal{C}$$

By Theorem 2.



Step 3: Constructing Colimits of Finitary Monads

A
Constructive
Small Object
Argument

Paul Seip

Introduction

Motivation

The Main
Theorem

Proof
Strategy

Conclusion

Theorem 3

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$$\begin{array}{ccc} \lim_{j \in \mathcal{J}} T_j\text{-Alg} & \xrightarrow{\cong} & T\text{-Alg} \\ & \searrow & \swarrow \\ & \mathcal{C} & \end{array}$$

The diagram shows a commutative triangle. The top-left node is $\lim_{j \in \mathcal{J}} T_j\text{-Alg}$, the top-right node is $T\text{-Alg}$, and the bottom node is $\mathcal{C}$. A blue arrow labeled $\cong$ points from $\lim_{j \in \mathcal{J}} T_j\text{-Alg}$ to $T\text{-Alg}$. A solid black arrow points from $\lim_{j \in \mathcal{J}} T_j\text{-Alg}$ down to $\mathcal{C}$. A solid black arrow points from $T\text{-Alg}$ down to $\mathcal{C}$. A dashed red arrow labeled T points from $\mathcal{C}$ back up to $\lim_{j \in \mathcal{J}} T_j\text{-Alg}$.

By Theorem 2.

By Beck's monadicity theorem.



Step 4: Finitary Small Object Argument for Categories

A

Constructive
Small Object
Argument

Paul Seip

Introduction

Motivation

The Main
Theorem

Proof
Strategy

Conclusion

Theorem 4

Let $\mathcal{C}$ be a locally small and cocomplete category and let $U : \mathcal{J} \rightarrow \mathcal{C}^2$ be a small category over $\mathcal{C}^2$ such that for every $j \in \mathcal{J}$ the domain and the codomain of Uj are ω -compact. Then the AWFS $(\mathcal{C}, L, R)$ cofibrantly generated by U exists and is finitary.

$$\begin{array}{ccc} \mathcal{J}^{\mathfrak{m}} & (P, \rho)\text{-Alg} & R\text{-Alg} \\ & \searrow & \\ & \mathcal{C}^2. & \end{array}$$



Step 4: Finitary Small Object Argument for Categories

A

Constructive
Small Object
Argument

Paul Seip

Introduction

Motivation

The Main
Theorem

Proof
Strategy

Conclusion

Theorem 4

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$$\begin{array}{ccc} \mathcal{J}^{\mathfrak{h}} & \xrightarrow{\cong} & (P, \rho)\text{-Alg} \\ & \searrow & \downarrow \\ & & \mathcal{C}^2. \end{array} \quad \text{R-Alg}$$



Step 4: Finitary Small Object Argument for Categories

A

Constructive
Small Object
Argument

Paul Seip

Introduction

Motivation

The Main
Theorem

Proof
Strategy

Conclusion

Theorem 4

Let $\mathcal{C}$ be a locally small and cocomplete category and let $U : \mathcal{J} \rightarrow \mathcal{C}^2$ be a small category over $\mathcal{C}^2$ such that for every $j \in \mathcal{J}$ the domain and the codomain of Uj are ω -compact. Then the AWFS $(\mathcal{C}, L, R)$ cofibrantly generated by U exists and is finitary.

$$\begin{array}{ccc} \mathcal{J}^{\mathfrak{h}} & \xrightarrow{\cong} & (P, \rho)\text{-Alg} \\ & \searrow & \uparrow \downarrow \\ & & \mathcal{C}^2 \end{array} \quad \text{R-Alg}$$

By Theorem 2,



Step 4: Finitary Small Object Argument for Categories

A

Constructive
Small Object
Argument

Paul Seip

Introduction

Motivation

The Main
Theorem

Proof
Strategy

Conclusion

Theorem 4

Let $\mathcal{C}$ be a locally small and cocomplete category and let $U : \mathcal{J} \rightarrow \mathcal{C}^2$ be a small category over $\mathcal{C}^2$ such that for every $j \in \mathcal{J}$ the domain and the codomain of Uj are ω -compact. Then the AWFS $(\mathcal{C}, L, R)$ cofibrantly generated by U exists and is finitary.

$$\begin{array}{ccccc} \mathcal{J}^{\mathfrak{m}} & \xrightarrow{\cong} & (P, \rho)\text{-Alg} & \xrightarrow[\cong]{} & R\text{-Alg} \\ & \searrow & \uparrow \dashv \downarrow & \swarrow & \\ & & \mathcal{C}^2 & & \end{array}$$

By Theorem 2, by Beck's monadicity theorem.



Step 4: Finitary Small Object Argument for Categories

A

Constructive
Small Object
Argument

Paul Seip

Introduction

Motivation

The Main
Theorem

Proof
Strategy

Conclusion

Theorem 4

Let $\mathcal{C}$ be a locally small and cocomplete category and let $U : \mathcal{J} \rightarrow \mathcal{C}^2$ be a small category over $\mathcal{C}^2$ such that for every $j \in \mathcal{J}$ the domain and the codomain of Uj are ω -compact. Then the AWFS $(\mathcal{C}, L, R)$ cofibrantly generated by U exists and is finitary.

$$\begin{array}{ccccc} \mathcal{J}^{\mathfrak{M}} & \xrightarrow{\cong} & (P, \rho)\text{-}\mathbf{Alg} & \xrightarrow{\cong} & \mathbf{R}\text{-}\mathbf{Alg} \\ & \searrow & \uparrow \downarrow & \swarrow & \\ & & \mathcal{C}^2 & & \end{array}$$

By Theorem 2, by Beck's monadicity theorem.

By Theorem 1, $\mathcal{J}^{\mathfrak{M}} \cong \mathbf{R}\text{-}\mathbf{Alg}$ for some AWFS (L, R) .



Step 5: Finitary Small Object Argument for Double Categories

A

Constructive
Small Object
Argument

Paul Seip

Introduction

Motivation

The Main
Theorem

Proof
Strategy

Conclusion

Theorem (finitary small object argument)

Let $\mathcal{C}$ be a locally small, cocomplete category, and let $U : \mathbb{J} \rightarrow \mathbf{Sq}(\mathcal{C})$ be a double functor subject to the following conditions

- 1** *$\mathbb{J}$ is small,*
- 2** *the object Uj is ω -compact for every object $j \in \mathcal{J}_0$.*

Then the AWFS cofibrantly generated by U exists and is finitary.



A

Constructive
Small Object
Argument

Paul Seip

Introduction

Motivation

The Main
Theorem

Proof
Strategy

Conclusion

$$(\mathbb{J}^{\mathfrak{M}})_1 \longrightarrow \mathcal{J}_1^{\mathfrak{M}} \xrightleftharpoons[(\delta_{\mathbb{J}})_1]{m^{\mathfrak{M}}} \mathcal{J}_2^{\mathfrak{M}}$$

R-Alg

R₁-Alg

R₂-Alg



A

Constructive
Small Object
Argument

Paul Seip

Introduction

Motivation

The Main
Theorem

Proof
Strategy

Conclusion

$$\begin{array}{ccccc} (\mathbb{J}^{\mathfrak{M}})_1 & \longrightarrow & \mathcal{J}_1^{\mathfrak{M}} & \xrightleftharpoons[(\delta_{\mathbb{J}})_1]{m^{\mathfrak{M}}} & \mathcal{J}_2^{\mathfrak{M}} \\ & & \downarrow \cong & & \downarrow \cong \\ \text{R-Alg} & & \text{R}_1\text{-Alg} & \xrightleftharpoons[t^*]{s^*} & \text{R}_2\text{-Alg} \end{array}$$

By Theorem 4



A

Constructive
Small Object
Argument

Paul Seip

Introduction

Motivation

The Main
Theorem

Proof
Strategy

Conclusion

$$\begin{array}{ccccc} (\mathbb{J}^{\mathfrak{M}})_1 & \longrightarrow & \mathcal{J}_1^{\mathfrak{M}} & \xrightleftharpoons[(\delta_{\mathbb{J}})_1]{m^{\mathfrak{M}}} & \mathcal{J}_2^{\mathfrak{M}} \\ & & \downarrow \cong & & \downarrow \cong \\ \mathbf{R}\text{-}\mathbf{Alg} & \xrightarrow{q^*} & \mathbf{R}_1\text{-}\mathbf{Alg} & \xrightleftharpoons[t^*]{s^*} & \mathbf{R}_2\text{-}\mathbf{Alg} \end{array}$$

By Theorem 4 , by Theorem 3.



A

Constructive Small Object Argument

Paul Seip

Introduction

Motivation

The Main
Theorem

Proof
Strategy

Conclusion

$$\begin{array}{ccccc}
 (\mathbb{J}^\natural)_1 & \longrightarrow & \mathcal{J}_1^\natural & \xrightleftharpoons[(\delta_{\mathbb{J}})_1]{m^\natural} & \mathcal{J}_2^\natural \\
 \cong \downarrow & & \cong \downarrow & & \downarrow \cong \\
 \mathbf{R}\text{-}\mathbf{Alg} & \xrightarrow{q^*} & \mathbf{R}_1\text{-}\mathbf{Alg} & \xrightleftharpoons[t^*]{s^*} & \mathbf{R}_2\text{-}\mathbf{Alg}
 \end{array}$$

By Theorem 4 , by Theorem 3.

By Theorem 1, $\mathbb{J}^\natural \cong \mathbf{R}\text{-}\mathbf{Alg}$ for some AWFS (L, R) .



Presheaf Categories

A

Constructive
Small Object
Argument

Paul Seip

Introduction

Motivation

The Main
Theorem

Proof
Strategy

Conclusion

Corollary 1

Let $\mathbb{C}$ be a small category and let $U : \mathbb{J} \rightarrow \mathbf{Sq}(\widehat{\mathbb{C}})$ be a double functor subject to the following conditions

- ① $\mathbb{J}$ is small,*
- ② Uj is finitely generated for every object $j \in \mathcal{J}_0$.*

Then the AWFS cofibrantly generated by U exists and is finitary.

Example: effective Kan fibrations.



Future Directions

A

Constructive
Small Object
Argument

Paul Seip

Introduction

Motivation

The Main
Theorem

Proof
Strategy

Conclusion

- Directly constructing the adjoint of $(\mathbb{J}^{\sharp})_1 \rightarrow \mathcal{C}^2$;
- Investigating whether the conditions of the main theorem can be relaxed, while maintaining a constructive proof;
- Proving a converse result for the main theorem;
- Implementing the proof in a theorem prover.



A
Constructive
Small Object
Argument

Paul Seip

Introduction

Motivation

The Main
Theorem

Proof
Strategy

Conclusion

-  Quillen, D.G. *Homotopical Algebra. Lecture Notes in Mathematics*, no. 43. Springer-Verlag, Berlin (1967)
-  Garner, R. *Understanding the small object argument*. Applied categorical structures, 17(3):247–285, 2009. arXiv:0712.0724.
-  Bourke, J. and Garner, R. *Algebraic weak factorisation systems I: Accessible AWFS*. Journal of Pure and Applied Algebra Volume 220, Issue 1, January 2016, Pages 108-147.
-  v/d Berg, B. and Faber, E. *Effective Kan Fibrations in Simplicial Sets*. Lecture Notes in Mathematics. Springer International Publishing, 2022.
-  Cohen et al. *Cubical Type Theory: A Constructive Interpretation of the Univalence Axiom*. TYPES 2015.